

I am Anany Sharma, an AI Engineer and Machine Learning researcher currently pursuing my Master's in Artificial Intelligence Systems at the University of Florida. My work focuses on building scalable AI systems that bridge research and production, with experience across large language models, Edge AI, distributed data systems, retrieval pipelines, and cloud infrastructure.

Over the past few years, I have worked across AI engineering, software development, and machine learning research in both industry and academic environments. At UnitedHealth Group, I built production AI and data systems involving retrieval augmented generation pipelines, large scale ETL workflows, and cloud based search infrastructure processing millions of records. I improved retrieval Recall@10 by 12 percent through semantic chunking, query rewriting, and ranking optimization while also automating over 1,500 recurring reporting workflows using distributed data pipelines.

As an NSF funded AI Research Fellow at the University of Florida, I currently work on efficient AI systems for constrained environments. I engineered a Small Language Model on Jetson Orin with a 96MB footprint using ONNX Runtime and TensorRT, achieving 70 tokens per second with 2.44ms time to first token through KV cache optimization, flash attention, and mixed precision techniques. I have also built Edge AI pipelines deployed on ESP32 devices and forecasting systems for environmental monitoring using temporal convolutional networks and attention based architectures.

My projects span multimodal AI, generative models, NLP, and agentic systems. I developed EMOGEN, a multilingual speech to image generation system integrating Whisper, CLIP guided diffusion models, and emotion recognition pipelines. I also built SMIRE AI, a multi agent healthcare assistant platform using LangChain, CrewAI, Pinecone, Flask, and Docker to support conversational retrieval and workflow automation.

Technically, I work across Python, PyTorch, TensorFlow, SQL, Spark, FastAPI, Docker, Kubernetes, AWS, Azure, Terraform, and modern LLM orchestration frameworks. I enjoy designing systems end to end, from model development and experimentation to scalable deployment, monitoring, and backend integration.

I am driven by building AI systems that are practical, efficient, and impactful. Whether working on production machine learning pipelines, Edge AI optimization, multimodal learning, or agentic workflows, I enjoy solving complex engineering problems and turning research ideas into deployable systems.